Product Development Canvas

Product: AgroVerify | Domain: Agriculture | Team: Agri_Developers | Date: Oct 2025 | Version: 1.0

Project: Agriculture — AgroVerify Input Verification System
Team: Agri_Developers

■ Purpose

What Is This Concept?

- AgroVerify is a mobile + IVR system that lets any Indian farmer verify the authenticity of seeds, fertilizers, and pesticides at point of purchase — before paying

Does It Solve a Problem?

- Yes — ~25-30% of pesticide market and 1-in-8 seed batches in India are fake/substandard, causing total crop failure with no recourse for the farmer

Is It Serving a Need or Creating One?

- Serving a critical, urgent, existing need — farmers already know fake inputs exist but have zero verification tool available to them today

Untapped Need It Taps

- Farmer-level access to government SATHI database — which exists but has no consumer-facing interface

■ Product Experience

What Should the Farmer Feel?

- Confidence — "I know this is genuine before I spend my money" | Safety — "I cannot be cheated on inputs anymore"
- Simplicity — verification in under 15 seconds, no tech knowledge needed | Empowerment — farmer has information the dealer does not expect
- Trust — in government system, in formal input market, in the crop they are about to sow

■ Product Functions

- **Verify** — scan or enter batch code → instant genuine/fake result (core function)
- **Alert** — suspicious batch detection auto-logged to agriculture officer dashboard
- **Inform** — show product details: manufacturer, expiry, certification status, registered dealer list
- **Report** — farmer can manually flag a suspicious product even without batch code (photo + location)
- **Educate** — push seasonal tips on what genuine inputs should look like for major crops

■ Customer Revalidation

How to Test With Users

- Pilot with 50 farmers across 5 Gujarat villages via KVK partnership — measure % who complete verification without assistance
- Record: time taken per verification, error rate, language preference, fallback to IVR

Key Questions to Validate

- Can a 55-year-old farmer with no smartphone use IVR fallback without help?
- Does offline cache work reliably in zero-connectivity villages?
- Do farmers trust the result enough to refuse to buy a SUSPECT product?

Success Metrics

- 90%+ farmers complete verification in under 60 seconds
- 70%+ would refuse purchase on SUSPECT result
- Zero false positives on GENUINE products in pilot

Revalidation Timeline

- After Phase 3 pilot (Month 4) — interview 20 farmers; shadow 10 purchase decisions

■ People

Key Customer Segment

- Small & marginal Indian farmers (<2 hectares) — 86% of all Indian farmers; buy inputs before every sowing season; cannot afford to lose a season to fake inputs

Profile

- Age 30-60; primary/secondary education; basic feature phone or low-end Android; Gujarati/Hindi/Marathi speaking; ■5,000-20,000 seasonal input budget

Secondary Segments

- Women farmers (perform most farm labor; equally vulnerable to fake inputs)
- State agriculture officers (need enforcement data)
- Agri-input dealers (need compliance tool to avoid selling fakes unknowingly)

■ Product Features

- QR Code Scanner — works in low-light shop conditions; offline-capable with cached batch data
- SMS/IVR Fallback — farmer dials toll-free number, enters batch code, hears result in local language
- Multilingual UI — Gujarati, Hindi, Marathi; result read aloud for low-literacy users
- Offline Mode — syncs SATHI data when connected; verifies from cache in zero-internet zones
- Dealer Compliance Dashboard — web portal for agriculture officers; maps fake-input detections by village/district
- Crowdsource Alert — 3+ suspect flags on same batch triggers automatic district-level alert

■■ Components

- SATHI API Connector — reads government batch registry; syncs daily; handles auth and rate limits
- QR Scanner Module — ML-Kit based; works on Android 6+; handles damaged/partial QR codes
- IVR Engine — Twilio or Exotel backend; toll-free number; DTMF batch-code entry; TTS result in 3 languages
- Offline Cache Layer — SQLite local DB; stores last 30-day batch data; auto-syncs on WiFi/4G
- Alert Engine — flags batch IDs with 3+ suspect reports; pushes to officer dashboard via REST API
- Officer Dashboard — React web app; heatmap of detections; exportable reports for enforcement action
- Analytics Backend — anonymized scan data; used to improve database coverage and detect fraud patterns

■ Reject / Redesign / Retain

Reject (Did Not Work in Testing)

- In-app payment for verification — farmers refused any paid feature; must be free
- Dealer-initiated verification — dealers feel targeted; farmer-initiated only

Redesign (Partially Useful)

- Camera-based label OCR — too slow and unreliable in poor light; redesign to manual batch-code entry as primary, OCR as secondary
- English-only result screen — redesigned to local language first with English secondary

Retain (Validated by Users)

- QR scan as primary verification — fast, accurate, preferred by smartphone users
- IVR/SMS fallback — critical for feature phone users; non-negotiable feature to retain
- Offline cache — farmers in low-connectivity areas depend on it; must retain and expand
- Instant result in local language — highest-rated feature in pilot feedback